

Basics of crystallography, part 2

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4.1 The description of crystal symmetry in International Tables A: Positions

Three different kinds of description of the symmetry operations of crystals are used in *International Tables A*:

- (1) By one or more diagrams of the symmetry elements, see Section 6.4.1.
- (2) By one diagram of 'general positions', see Section 6.4.4.
- (3) By the coordinate triplets of the 'general positions', see Section 3.1.1. As shown there, the coordinate triplets not only specify the coordinates of the image points, but can also be regarded as *descriptions of the mappings*; see also Section 6.4.3.

In *International Tables A*, the coordinate triplets in the upper block of the 'Positions', such as shown at the beginning of Section 3.1.1 for the space group $I4_1/amd$, are a kind of shorthand notation of eqn (3.1) (page 20):

- the left part ($\bar{x} =, \bar{y} =, \bar{z} =$) has been omitted;
- all components with the coefficients $W_{ik} = 0$ and $w_i = 0$ have been omitted.

The term (2) $\bar{x} + \frac{1}{2}, \bar{y} + \frac{1}{2}, \bar{z} + \frac{1}{2}$ therefore means:

$$W = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad w = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

The term (3) $\bar{y}, x + \frac{1}{2}, \bar{z} + \frac{1}{4}$ is a shorthand notation of the matrix-column pair:

$$W = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad w = \begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{4} \end{pmatrix}$$

This way, *International Tables A* present the analytic-geometrical tools for the description of crystal symmetry.

4.2 Crystallographic symmetry operations

By definition, crystallographic symmetry operations are always isometries; however, not every isometry can be a crystallographic symmetry operation.

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This is due to the periodicity of crystals and to the restriction that the periodicity lengths may not be arbitrarily short.

Let a crystal be referred to a coordinate system with a primitive basis, Definition 2.7 (page 14). Then every symmetry operation corresponds to a matrix–column pair (W, w) . This includes an infinity of translations (I, w) ; every triplet w of integers represents a translation. However, the number of matrix pairs W is finite, as follows from the following considerations.

Theorem 4.1 For every space group, represented by the matrix–column pairs (W, w) of its symmetry operations, there exist only a finite number of matrices W .

The matrix W maps the basis (a_1, a_2, a_3) onto the vectors $(\bar{a}_1, \bar{a}_2, \bar{a}_3)$. For the set of all W s we have a set of base images. Every basis vector is a lattice vector; therefore, its image vector is also a lattice vector, since the lattice has to be mapped onto itself under W , otherwise W would not correspond to a symmetry operation. The set of all image vectors of a basis vector, say a_i , have their endpoints on a sphere of radius a_i . If there were infinitely many image vectors $\bar{a}_i$, their endpoints would have to concentrate around at least one point on this sphere with an infinite density. For any two lattice vectors their difference also is a lattice vector; then there would have to exist arbitrarily short lattice vectors. As a consequence, only a finite number of image vectors $\bar{a}_i$ can exist for any i , and thus only finitely many matrices W .

This conclusion is evidently independent of the dimension d of space. However, the maximal possible number of different matrices increases markedly with d : It is 2 for $d = 1$, 12 for $d = 2$, 48 for $d = 3$, and 1152 for $d = 4$.

A second restriction concerns the possible orders N of rotations. According to the aforementioned consideration of the matrices W , these have to consist of integral numbers if a primitive basis has been chosen, because in that case all lattice vectors are integral. On the other hand, the matrix of any rotation, referred to an appropriate orthonormal basis, can be represented by

$$W = \begin{pmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The trace of the matrix is therefore $\text{tr}(W) = n$, n integral, as well as $\text{tr}(W) = 1 + 2 \cos \varphi$.

The trace is independent of the basis. As a consequence, for crystallographic symmetry operations, one has:

$$1 + 2 \cos \varphi = n, \quad n \text{ integral}$$

This restricts the possible values of φ to:

$$\varphi = 0^\circ, 60^\circ, 90^\circ, 120^\circ, 180^\circ, 240^\circ, 270^\circ, \text{ and } 300^\circ$$

Therefore, the order N is restricted to the values $N = 1, 2, 3, 4$, and 6. This is also valid for rotoinversions, since any rotation can be coupled with an inversion.

In crystallography, the types of crystallographic symmetry operations are designated by their **Hermann-Mauguin symbols** (Figs. 4.1 and 4.2). These are:

- 1 for the identical mapping.
- $\bar{1}$ ('one bar') for the inversion.
- Rotations: A number N , $N = 2, 3, 4, 6$. This corresponds to the order of the rotation. If needed, the power of the rotation is mentioned; for example, $6^{-1} = 6^5$, rotation by $-60^\circ = 300^\circ$.
- Screw rotations: N_p designates a screw rotation consisting of a rotation N coupled with a translation parallel to the axis of rotation by p/N of the shortest lattice distance in this direction. The possible symbols are: 2_1 ('two sub one'), 3_1 , 3_2 , 4_1 , 4_2 , 4_3 , 6_1 , 6_2 , 6_3 , 6_4 , and 6_5 .
- Rotoinversions: $\bar{3}$, $\bar{4}$ and $\bar{6}$.
- Reflections: m (like *mirror*). m is identical to $\bar{2}$.

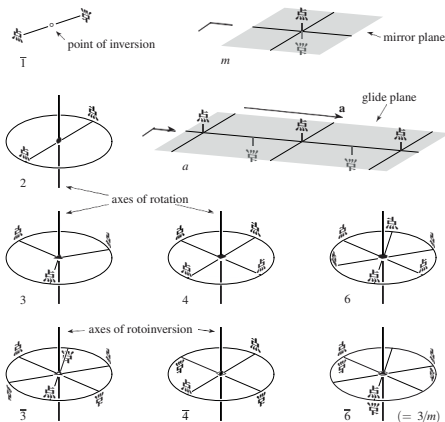


Fig. 4.1 The effect of different symmetry operations on the point 点 (Chinese symbol for point, pronounced diǎn in Chinese, hoshee in Japanese). The symmetry operations are designated by their Hermann-Mauguin symbols and by their graphical symbols.

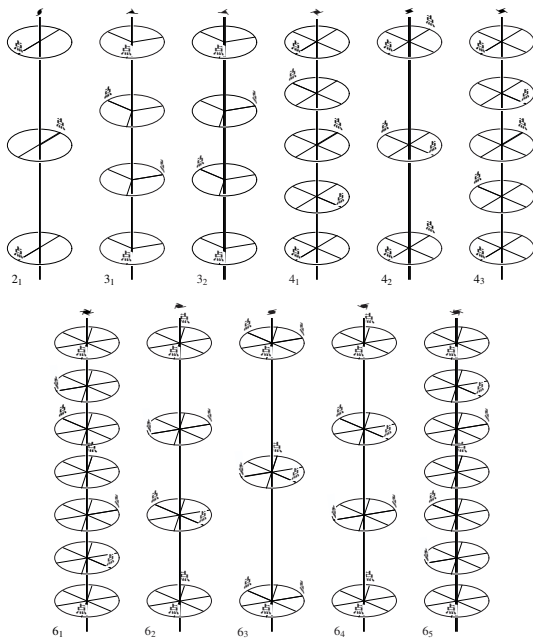


Fig. 4.2 The crystallographic screw axes with their Hermann-Mauguin and graphical symbols. The axes 3_1 , 4_1 , 6_1 , and 6_2 are right-handed, 3_2 , 4_3 , 6_3 , and 6_4 left-handed. One length of translation in the axis direction is depicted.

- Glide reflections: The letter m is replaced by a symbol that expresses the glide vector $\mathbf{v}$. The vector $\mathbf{v}$ is parallel to the glide plane; its length is one-half of a lattice vector. The main symbols among the conventional settings are:

a , b , or c if the glide vector is $\frac{1}{2}\mathbf{a}$, $\frac{1}{2}\mathbf{b}$, or $\frac{1}{2}\mathbf{c}$, respectively; for plane groups it is g ;

n for glide vectors $\frac{1}{2}(\pm\mathbf{a} \pm \mathbf{b})$, $\frac{1}{2}(\pm\mathbf{b} \pm \mathbf{c})$, $\frac{1}{2}(\pm\mathbf{c} \pm \mathbf{a})$, or $\frac{1}{2}(\pm\mathbf{a} \pm \mathbf{b} \pm \mathbf{c})$;

d for glide vectors $\frac{1}{4}(\pm\mathbf{a} \pm \mathbf{b})$, $\frac{1}{4}(\pm\mathbf{b} \pm \mathbf{c})$, $\frac{1}{4}(\pm\mathbf{c} \pm \mathbf{a})$, or $\frac{1}{4}(\pm\mathbf{a} \pm \mathbf{b} \pm \mathbf{c})$.

e designates a glide plane having two mutually perpendicular glide vectors $\frac{1}{2}\mathbf{a}$, $\frac{1}{2}\mathbf{b}$, or $\frac{1}{2}\mathbf{c}$.

For the special symbols g_1 and g_2 among non-conventional settings of tetragonal space groups see Section 9.3.3.

The mentioned symbols for single crystallographic symmetry operations give no indication about the orientation of the rotation or rotoinversion axis. As explained in more detail in Section 6.3.1, the orientation of an axis is expressed in a Hermann–Mauguin symbol by its position in the symbol.

In the plane, the same orders exist for rotations, since the traces of the matrices yield the same equation $2\cos\varphi = n$ with the same solutions for φ and N . However, only one type of symmetry operation exists in the plane for $\det(\mathbf{W}) = -1$, the reflection or glide reflection at a line, which is represented in an appropriate basis by:

$$\mathbf{W} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

A rotation axis belongs to every rotation. Among the symmetry rotations of crystals, these axes can only adopt certain mutual angles, otherwise their compositions would produce rotations with non-integral traces. This is another way to understand why for a crystal there can only exist a finite number of matrices $\mathbf{W}$. The different possible sets $\{\mathbf{W}\}$ of compatible matrices $\mathbf{W}$, the 32 crystal classes, can be derived in this way.

4.3 Geometric interpretation of the matrix–column pair ($\mathbf{W}, \mathbf{w}$) of a crystallographic symmetry operation

Consider a crystallographic symmetry operation W , represented by the matrix–column pair $(\mathbf{W}, \mathbf{w})$, referred to a given coordinate system (the geometric interpretation would be impossible without the coordinate system).

Essentially, the following procedure can be applied to general isometries, without restrictions to the orders. See also Section 3.6.

First, we analyse the matrix part $\mathbf{W}$:

- $\det(\mathbf{W}) = +1$: rotation; $\det(\mathbf{W}) = -1$: rotoinversion;
- angle of rotation φ from $\cos\varphi = \frac{1}{2}(\pm\text{tr}(\mathbf{W}) - 1)$ (4.1);
the $+$ sign refers to rotations, the $-$ sign to rotoinversions.

This can be summarized in a table:

	$\det(W) = +1$					$\det(W) = -1$				
$\text{tr}(W)$	3	2	1	0	-1	-3	-2	-1	0	1
type	1	6	4	3	2	$\bar{1}$	$\bar{6}$	$\bar{4}$	$\bar{3}$	$\bar{2} = m$
order	1	6	4	3	2	2	6	4	6	2

Characterization of the crystallographic symmetry operations

Every symmetry operation, translations excepted, is related to a *symmetry element*. This is a point, a straight line, or a plane which retains its position in space when the symmetry operation is performed.

1. Type 1 or $\bar{1}$:

- 1: identity or translation with $\mathbf{w}$ as a column of the translation vector;
- $\bar{1}$: inversion; the symmetry element is the *point of inversion* F (inversion centre):

$$\mathbf{x}_F = \frac{1}{2}\mathbf{w} \quad (4.2)$$

2. All other operations have a fixed axis (axis of rotation or rotoinversion); its direction can be calculated from $\mathbf{W}\mathbf{u} = \mathbf{u}$ (rotations) or $\mathbf{W}\mathbf{u} = -\mathbf{u}$ (rotoinversions). For a reflection and a glide reflection the symmetry element is not the axis, but the mirror plane or the glide plane (short for glide-reflection plane); the direction of the axis is normal to the plane.
3. The *coefficient of a screw rotation* or the *coefficient of a glide reflection* $\frac{1}{k}\mathbf{t}$ can be calculated from the corresponding matrix of rotation $\mathbf{W}$, which has order k , i.e. $\mathbf{W}^k = \mathbf{I}$:

$$\frac{1}{k}\mathbf{t} = \frac{1}{k} \left(\mathbf{W}^{k-1} + \mathbf{W}^{k-2} + \dots + \mathbf{W} + \mathbf{I} \right) \mathbf{w} \quad (4.3)$$

If $\mathbf{t} = \mathbf{o}$, we have a rotation or a reflection. If $\mathbf{t} \neq \mathbf{o}$, we have a screw rotation or a glide reflection. In this case one obtains the *reduced operation*:

$$\left(\mathbf{I} - \frac{1}{k}\mathbf{t} \right) (\mathbf{W}, \mathbf{w}) = \left(\mathbf{W}, \mathbf{w} - \frac{1}{k}\mathbf{t} \right) = (\mathbf{W}, \mathbf{w}') \quad (4.4)$$

The column $\frac{1}{k}\mathbf{t}$ is called the *screw component* or the *glide component* of the column $\mathbf{w}$. The column $\mathbf{w}' = \mathbf{w} - \frac{1}{k}\mathbf{t}$ determines the *position* in space of the corresponding symmetry element. Therefore, $\mathbf{w}'$ is also called the positional component of $\mathbf{w}$. If $\mathbf{W}$ has only main diagonal coefficients, i.e. only $W_{ii} \neq 0$, then $W_{ii} = \pm 1$ holds and $\mathbf{w}_i$ is a screw or a glide coefficient for $W_{ii} = +1$ and a positional coefficient for $W_{ii} = -1$.

4. The fixed points of the isometry are determined by solution of eqn (3.11):

$$\mathbf{W}\mathbf{x}_F + \mathbf{w} = \mathbf{x}_F$$

This equation has no solution for screw rotations and glide reflections. The position of the screw axis or glide plane rather results from the reduced operation, eqn (4.4):

$$\mathbf{W}\mathbf{x}_F + \mathbf{w}' = \mathbf{x}_F \quad (4.5)$$

The conventional pairs (W, w) are listed in *International Tables A* in short-hand notation as ‘general positions’, see Section 3.1.1. Their geometric meaning can be found in the tables of the space groups under the heading ‘symmetry operations’. They have been numbered in the same sequence as the coordinate triplets. More explanations follow in Section 6.4.3.

4.4 Derivation of the matrix–column pair of an isometry

The matrix–column pair (W, w) consists of 12 coefficients. To determine them, the coordinates of four non-coplanar image points have to be known. The most straightforward procedure is to take the image points of the origin and of the three endpoints of the basis vectors.

1. If $\bar{O}$ is the image point of the origin O , we obtain:

$$\bar{o} = W\mathbf{o} + \mathbf{w} = \mathbf{w} \quad (4.6)$$

Therefore, the coordinates $\bar{o}$ of $\bar{O}$ are the coefficients of $\mathbf{w}$.

2. After having determined $\mathbf{w}$, the matrix W is obtained from the images of the points X_o, Y_o and Z_o with

$$\mathbf{x}_o = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{y}_o = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{z}_o = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

using the relations:

$$\bar{\mathbf{x}}_o = W\mathbf{x}_o + \mathbf{w}, \quad \bar{\mathbf{y}}_o = W\mathbf{y}_o + \mathbf{w}, \quad \bar{\mathbf{z}}_o = W\mathbf{z}_o + \mathbf{w} \quad \text{or} \quad (4.7)$$

$$\bar{\mathbf{x}}_o = \begin{pmatrix} W_{11} \\ W_{21} \\ W_{31} \end{pmatrix} + \mathbf{w}, \quad \bar{\mathbf{y}}_o = \begin{pmatrix} W_{12} \\ W_{22} \\ W_{32} \end{pmatrix} + \mathbf{w}, \quad \bar{\mathbf{z}}_o = \begin{pmatrix} W_{13} \\ W_{23} \\ W_{33} \end{pmatrix} + \mathbf{w} \quad (4.8)$$

With the coordinates of $\bar{\mathbf{x}}_o, \bar{\mathbf{y}}_o, \bar{\mathbf{z}}_o$ one obtains the matrix W .

To check the result, one calculates fixed points, the trace, the determinant, the order, and/or other known values. If the images of $\bar{O}, \bar{X}_o, \bar{Y}_o$, or $\bar{Z}_o$ are difficult to determine, one has to use the images of other points; the calculations then become more complicated.

Exercises

Solutions in Appendix D (page 283)

- (4.1) Give the geometric interpretations for the symmetry operations (8) , (10) , (15) , $(15) + (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}) = (15)_2$ and $(15)_{2n}$ mentioned in Exercises 3.1 and 3.2 (page 40; $(15)_{2n}$ means standardized). Take the matrices you obtained with Exercise 3.1 and apply the procedure explained in Section 4.3. For each of the mentioned operations, derive:
- (a) the determinant $\det(W)$ and the trace $\text{tr}(W)$;
 - (b) from that, the type of symmetry operation;

(c) the direction of the axis of rotation or of the normal to the plane;

(d) the screw and glide components;

(e) the position of the symmetry element;

(f) the Hermann–Mauguin symbol of the symmetry operation.

(g) Which operations yield fixed points?

Compare the results with the figures and listings of the symmetry operations in *International Tables A*, space group $I4_1/amd$, origin choice 2.